[INCH-POUND]
A-A-55477
August 4, 1995

COMMERCIAL ITEM DESCRIPTION

MULTIMETER, HAND HELD, 3 2/3 DIGIT, COMBINATION MULTIMETER/OSCILLOSCOPE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies

1 SCOPE This CID covers the general requirements for a hand held, combination multimeter/oscilloscope which provides the functionality of a 3 2/3 digit, digital multimeter with greater than 3000 counts resolution and a 50 MHz, 25 mega-samples per second sample rate, dual-channel digital storage oscilloscope with memory to store and recall 8 waveforms. For the purposes of this CID, multimeter/oscilloscope are referred to herein as the instrument This instrument shall be capable of simultaneous display of the input waveform and the selected multimeter function Instruments covered by this CID are intended for commercial/industrial applications.

2 CLASSIFICATION This CID uses a classification system which is included in the Part or Identification Number (PIN) as shown in the following example (see 6.1)

AA55477 - 001

AA55477: CID number
001: Dash number

3 SALIENT CHARACTERISTICS

3.1 Design and construction Instruments supplied to this CID shall be as specified herein and shall meet the requirements of MIL-T-28800 for type III, class 3, style E, equipment, category III of IEC 1010-1, and UL 1244.

3.2 Safety This instrument shall be designed to category III of IEC 1010-1 and designed and listed to UL 1244 for 600 volts (V) rms measurements on industrial power distribution circuits.

3.3 Maximum voltage isolation to earth The maximum voltage isolation to earth shall be 600 V rms from any terminal.

3.4 Display The useable display area shall include at least eight vertical and eight horizontal (ten if cursor not used) divisions, with internal graticule ruled in squares at least 8.9 millimeters (mm) (.35 inch) in size Display resolution shall be at least 240 x 240 pixels, 25 pixels per division. An liquid crystal display (LCD) shall have selectable backlighting A portion of the display area shall be useable for simultaneous indication of inout waveform and multimeter reading

3.5 Instrument dimensions This instrument shall have a maximum height of 60 mm (2.4 inches), a maximum width of 130 mm (5.1 inches), and a maximum length of 260 mm (10.2 inches) A maximum height of 65 mm (2.5 inches), a maximum width of 140 mm (5.5 inches), and a maximum length of 275 mm (10.8 inches) with the holster.

3.6 Weight The weight of this instrument shall be 1.8 kilograms (4.0 pounds) maximum with holster and 1.5 kilograms (3.3 pounds) maximum without holster

3.7 Protective holster A shock-absorbing protective holster shall be provided with this instrument

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, or telephone (513) 296-6189, or facsimile (FAX) (513) 296-8868.

AMSC N/A FSC 6625

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3.8 Oscilloscope probes. This instrument shall be supplied with a set (2) of 10:1 oscilloscope probes. These probes shall be 1.5 meters (5 feet) in length and rated at 600 V rms, (one red and one grey in color). Scope and multimeter voltage measurements shall be made using the same (common) probe

3 9 Power source This instrument shall operate from nominal ac power, and shall also include internal chargeable batteries and charger to provide at least 4 hours operation following a full charge cycle (16 hours)

3 10 Optically isolated interface This instrument shall be supplied with an optically isolated RS232 interface for remote capabilities

3.11 Serial printer output This instrument shall be supplied with a serial printer output.

3 12 Environment.

3 12.1 Operating temperature This instrument shall perform to its full specified accuracy from +18° Celsius (C) to +28°C (+64 4° fahrenheit (F) to 82 4°F), and all ranges and functions from 0°C to +50°C (+32.0°F to 122.0°F)

3 12.2 Storage temperature. The storage temperature of this instrument shall be -20°C to +70°C (-4.0°F to +158.0°F).

3 12.3 Operating humidity The operating humidity for the temperature range of +20°C to +30°C (+68.0°F to +86 0°F) shall be 90 percent relative humidity (RH) (noncondensing) and for the temperature range of +30°C to +50°C (+86 0°F to +122.0°F) shall be 70 percent RH (noncondensing)

3 12 4 Storage humidity The storage humidity shall be 95 percent relative humidity maximum

3 12.5 Operating altitude The operating altitude shall be 3 kilometers (km) (10,000 feet) maximum

3.12.6 Storage altitude The storage altitude shall be 12 km (40,000 feet) maximum

3 12.7 Shock and vibration The shock and vibration requirements shall be in accordance with MIL-T-28800 for a style E, class 3 instrument

3 13 Signal generator This instrument shall include a signal output providing selectable sinewave and squarewave outputs, and voltage and current ramp for circuit and component testing

3 13.1 Sinewave The signal generator shall be able to produce a sinewave that is 1 V peak-to-peak at 976 Hz with less than 3 percent distortion

3 13.2 Squarewave The signal generator shall be able to produce a 5 V peak-to-peak (±2.5 V) squarewave at 488 Hz, 976 Hz, or 1 95 kHz

3 13.3 Component tester

3 13.3 1 Current mode. This instrument shall be able to produce a 4 second current ramp from 0 to +3 milliampere (mA) in 128 discrete steps The voltage compliance shall be less than or equal to 2 V

3 13.3.2 Voltage mode This instrument shall be able to produce a 4 second voltage ramp from -2 V to +2 V in 128 discrete steps The current compliance shall be ±1 mA

3 14 Oscilloscope function

3 14 1 Vertical inputs This instrument shall have two independent vertical inputs, with input common isolated from the instrument's chassis to protect against electrical shock

3 14.2 Analog bandwidth. The analog bandwidth of the vertical inputs shall be dc to 50 MHz.

3.14.3 Sample rate. This instrument shall be capable of sampling at 25 mega-samples per second.

3.14.4 Automatic setup. A single button control shall be provided that initiates automatic adjustment of the vertical and horizontal deflection factors and trigger level, as appropriate, to provide an optimized display of the input signal This function shall operate for signals exceeding 1 percent duty cycle and 50 Hz frequency.

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3.14.5 Vertical.

3 14 5 1 Vertical sensitivity. The deflection sensitivity shall be 1 millivolt (mV) per division to 100 V per division. 200 V, 500 V, or 1 kilovolt (kV) per division using the 10:1 probe

3 14.5.2 Accuracy The accuracy over eight major graticule divisions shall be ± (2 percent + 1 pixel)

3 14.5.3 Maximum input voltage. The maximum input voltage shall be 300 V (dc + peak ac), and 600 V (dc + peak ac) when using 10 1 probe

3 14.5.4 Vertical modes This instrument shall be able to display channel A, channel B, channel A&B inverted, channel A + channel B, channel A channel B, or channel A = y, and channel B = x

3.14.6 Horizontal

3 14 6 1 Time Base

3 14 6.1 1 Range Time base range shall be from 10 nanoseconds (ns) per division to 60 seconds per division.

3.14.6.1.2 Accuracy Time base accuracy shall be ±(0.1 percent + 1 pixel) of display over eight major graticule divisions. (1 pixel is equal to 1/25 of a division.)

3 14 6 2 Triggering Internal and external triggering shall be provided

3 14 6.2 1 Sensitivity Trigger sensitivity for internal mode shall be less than 0.8 divisions to 10 MHz, degrading to less than 1 5 divisions to 60 MHz, and less than 4 divisions to 100 MHz External triggering shall be +0.2 V or +2 V level transistor-transistor logic (TTL) compatible

3.14.7 Parametric measurements This instrument shall provide cursors that allow measurement of delta volts, delta time, frequency, 1/delta time, maximum, minimum, peak-to-peak, rms, average, phase, and rise and fall time Markers shall be provided to indicate points on the waveform for peak to peak, maximum peak, minimum peak, frequency, rise time, or phase measurements.

3 14.7.1 AC measurement This instrument shall respond to and measure the true rms value of ac signals

3 14.8 Waveform storage This instrument shall be able to store and recall eight waveforms

3 14.9 Set-up memory This instrument shall be able to store and recall ten front panel set-ups.

3 15 Multimeter function.

3 15.1 Multimeter display. The multimeter display shall be 3 2/3 digits having a digital readout of greater than 3,000 counts resolution

3.15.2 Memory The multimeter function shall be able to capture and hold for display, and store and display, the maximum, minimum, and average of all readings taken over an indefinite period of time. An audible tone shall indicate when a new minimum or maximum value has been recorded It shall be possible to display the difference between a reading stored in memory and any subsequent readings

3.15.3 Indicators Overload and polarity (+ and -) indicators shall be provided

3.15.4 Multimeter input. Input connectors for the multimeter function shall include banana jacks spaced 19.05 mm (.75 inch) apart, to accommodate dual-banana connectors and BNC female connector

3.15.5 DC/ac voltage, ohms, and frequency accuracy The accuracy is specified as ± ([percentage of reading] ± [number of least significant digits]) at +18°C to +28°C (+64°F to +82°F) with relative humidity to 90 percent for a period of 1 year after calibration. For temperatures less than +18°C or greater than +28°C, 0.1 x specified accuracy shall be added.

3.15 6 Voltage measurement. This instrument shall provide the voltage range, resolution, and accuracy as follows.

3.15.6.1 Range This instrument shall be capable of measuring 300 mV to 600 V full scale

3 15 6 2 Resolution The maximum resolution shall be 0 1 mV

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3 15.6.3 Accuracy The accuracy shall be ±(0.5 percent of reading + 5 digits) for dc readings; ±(2 percent of reading + 15 digits) for ac readings from 50 Hz to 20 kHz and 0 to 600 V rms.

3 15.6 4 Units The measured value shall be displayed in selected units of volts, dBV, or dBm referenced to 30 V, 75 V, 93 V, 135 V, 300 V, 600 V, or 1000 V

3 15 6 5 Noise rejection The dc common mode rejection shall be greater than 100 decibels (dB) at dc, 50 Hz, 60 Hz, and 400 Hz The ac common mode rejection shall be greater than 60 dB at dc to 60 Hz

3 15 6.6 Crest factor This instrument shall perform at up to 3.0 crest factor

3 15 7 Resistance measurements This instrument shall provide the resistance range, resolution, and accuracy as follows

3 15 7 1 Ranges This instrument shall be capable of measuring 0 to 30 megohms in seven ranges

3.15 7.2 Resolution The maximum resolution shall be 0 1 ohm

3 15 7.3 Accuracy The accuracy shall be ±(2 5 percent + 25 digits) for the 30 ohm range and ±(0.5 percent + 5 digits) for all other ranges

3 15.7 4 Overload protection This instrument shall be capable of continuously withstanding, without damage, 600 V rms using the 10 1 probe

3 15.8 Frequency measurement This instrument shall provide the frequency range, resolution, and accuracy as follows.

3 15 8 1 Range This instrument shall be capable of measuring 1 Hz to 5 MHz, pulse width 02 microsecond

3 15.8.2 Resolution. The maximum resolution shall be 0.01 Hz

3.15.8.3 Accuracy The accuracy shall be ±(0.5 percent of reading + 2 digits).

3 15.9 Diode test This instrument shall be capable of displaying diode or transistor junction voltage drops to at least 4 V A momentary audible tone shall indicate a forward biased junction, and the absence of sound shall indicate a reversed biased or open junction

3.15 9.1 Maximum test current The maximum diode or continuity test current shall be 500 microamperes (μA)

3.15.9.2 Maximum full scale voltage The maximum full scale voltage shall be 2.8 V.

3 16 Technical documentation Instruments shall be provided with technical documentation including owner's manual, and related information on operation and use

3.17 Marking Instruments supplied to this CID shall be marked with the manufacturer's standard commercial PIN and the UL 1244 listing mark.

4 QUALITY ASSURANCE PROVISIONS

4.1 Contractor certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

4.2 Certificate of compliance Each contractor desiring to be listed as a suggested source of supply shall submit a DESC certificate of compliance to DESC-ELDM. This certificate shall state that the contractor's product meets all the requirements of MIL-T-28800 for type III, class 3, style E, equipment; UL 1244; category III of IEC 1010, and all the requirements of this CID. In addition, a certificate of compliance shall accompany all instruments supplied to this CID.

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4.3 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID

a The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years

b The company must have sold 1,000 units meeting this CID in the commercial market place over the past 2 years

5 PACKAGING

5.1 Preservation, packing, and marking Preservation, packing, and marking shall be as specified in the contract or order.

6 NOTES

6.1 PIN The PIN should be used for Government purposes to buy commercial products to this CID See section 2 for PIN format example.

6.2 CAGE code For ordering purposes, inventory control, and submission of these instruments to DESC under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

6.3 Source of documents

Military Specification

MIL-T-28800 - Test Equipment For Use With Electrical And Electronic Equipment, General Specification For

(Copies of military specification are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 1010-1 - Safety Requirements For Electrical Equipment For Measurement, Control, and Laboratory Use.

(Applications for copies should be addressed to the International Electrotechnical Commission, 1 Rue De Varembe', Geneve, Switzerland)

UNDERWRITERS LABORATORIES, INCORPORATED (UL)

UL 1244 - Electrical and Electronic Measuring and Testing Equipment, Safety Standard For

(Applications for copies should be addressed to the Underwriters Laboratories, Incorporated, 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Industry association specifications and standards are generally available for reference from libraries They are also distributed among technical groups and using Federal agencies.)

6.4 Government users To acquire information on obtaining these instruments from the Government inventory system, please contact Defense Electronics Supply Center, ATTN DESC-EOE, 1507 Wilmington Pike, Dayton, OH 45444-5521, or telephone (513) 296-5168 and ask for AA55477-001

6.5 Ordering data The contract or order should specify the following

a CID document number, revision, and CID PIN

b. Quality assurance provisions

c. If a bid sample is required.

d. Packaging requirements.

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6 6 Suggested source of supply A suggested source of supply is listed below Additional sources will be added as they become available

AA55477	Vendor commercial PIN CAGE 89536
001	Fluke-97

Vendor CAGE number	Vendor name and address
89536	Fluke Corporation P O Box 9090 Everett, WA 98206-9090 (206) 356-5443

CIVIL AGENCY COORDINATING ACTIVITY

GSA - 7FXE

PREPARING ACTIVITY

DLA-ES

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